

# Maintenance & Workshop Team

## Detailed System Role & Workflow

Maintenance teams handle preventive maintenance, repairs, spare parts, and breakdown management.

### 1. Primary Responsibilities

- Perform preventive maintenance
- Handle emergency repairs
- Track spare parts
- Create maintenance reports
- Monitor workshop operations

### 2. Full Operational Flow

Maintenance teams receive alerts from:

- Mileage thresholds
- Breakdown reports
- Device failures
- Operational inspections

The system tracks:

- Workshop tasks
- Spare parts
- Downtime
- Maintenance history
- Bus readiness

### 3. System Interactions

- Interacts with ERP modules
- Interacts with live operational data
- Uses dashboards and reports
- Triggers workflow actions
- Generates system events
- Consumes real-time information
- Affects operational flow and business logic

## 4. Real-Time Data Impact

This role directly interacts with the real-time transportation ecosystem.

Actions performed by this role can affect:

- Bus operations
- Driver assignments
- Camera monitoring
- GPS tracking
- Financial workflows
- Maintenance workflows
- AI safety alerts
- Passenger information systems

Because the platform is interconnected, actions performed by one role automatically affect other departments and operational flows.

## 5. Important Developer Considerations

- Role-based access control (RBAC) is critical
- Permissions must be modular and scalable
- Realtime updates are required
- Audit logs should track all actions
- Cross-department workflow integration is necessary
- Dashboard views should differ by role
- Notification systems should be role-aware
- Operational events should be linked across modules

## 6. Final Assessment

The Maintenance & Workshop Team role is a critical part of the SEUM intelligent transportation ecosystem.

This role interacts not only with traditional ERP workflows, but also with:

- Real-time transportation operations
- AI safety systems
- GPS tracking infrastructure
- Smart bus devices
- Live camera monitoring
- Financial and operational analytics

The system must therefore be designed with interconnected workflows and scalable permission architecture.